# Project 2

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| Book: | Project 2 | | |

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## 1. Project

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| **Faculty:** | Information Technology |
| **Module Code:** | ITSCA2-12 |
| **Module Name:** | Scientific Computing with Python |
| **Content Writer:** | Dr Yves Matanga |
| **Internal Moderation:** | Community of Practice |
| **Copy Editor:** | Rebecca van Besouw |
| **Total Marks:** | 100 |
| **Submission Week:** | Second Block, Week 6 |

This module is presented on NQF level 6.

5% will be deducted from the student's project mark for each calendar day the project is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual project.

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**This project contributes 40% towards the final mark.**

[1] Under no circumstances will projects be accepted for marking after the projects of other students have been marked and returned to the students.

## 2. AI Checklist and Declaration

Before you submit an assignment, you should be able to confidently and honestly make all the below statements. For group work, you can also review the list, together, to hold one another accountable.

- I confirm that my submission reflects my personal learning, knowledge, skills, and understanding.
- If AI tools were employed for generating any part of this assignment (even in the drafting/research phase), I have referenced the use of AI in the text and/or declared the use of AI. I am willing to discuss the process and its contribution to my learning.
- I am aware that the lecturer may request a demonstration of my learning, such as explaining choices in approach, research, and the content I am submitting.
- I am aware that, if I did use AI in any phase of preparing this submitted work, it is recommended that I save a copy of the relevant chat history (prompts and answers), as this will help me demonstrate my writing/work process to my lecturer, if I am asked to do so.
- I have read the assignment instructions on whether AI tools are prohibited for this assignment, and if they are prohibited, I can confirm that I did not use AI tools.
- I understand that failure to agree to these terms may be deemed unethical, potentially leading to disciplinary action. I understand my responsibility for the integrity of my work, including seeking clarification from academic staff and adhering to instructions.

It is essential to acknowledge your use of ChatGPT or other generative AI in your learning. If you use ChatGPT or other generative AI to help you generate ideas or plan your process, you should still acknowledge how you used the tool, even if you don't include any AI-generated content in the assignment.

**Please note:** The following guiding questions that you will be asked in an AI declaration questionnaire below this assignment brief.

### AI Declaration

**It is compulsory to complete this AI declaration for each of your assignment submissions.**

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| I carefully read the assignment instructions, and the extent to which AI may be used for the assignment. |
| I used the following AI system(s)/tool(s): |
| I used it for the following: |

| |
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| If I quoted or paraphrased an AI output, I have referenced the relevant tool, version, and the date I used the tool. |
| I still consider this work my own. (i.e., I have not outsourced the final product, or significant portions of it, to AI tools/systems). |
| If required, I can defend my argument/perspective, explain my choices and approach, and can show that I am knowledgeable about the details of my work. |

For further guidance on the use of AI at Eduvos, please refer to the AI FAQ glossary. You will locate the FAQs in the Artificial Intelligence tile on the myDocuments page of myLMS.

# 3. Question 1

## Question 1 25 Marks

Study the scenario and complete the question(s) that follow:

**Mutuka Automotive Car Reseller**

Mutuka Automotive is a well-established car reseller with over a decade of experience in the pre-owned vehicle market. The company acquires and resells vehicles from previous owners and has historically relied on a manual valuation process supported by accumulated pricing knowledge in its historical records.

For this project, the supplied dataset contains vehicle specification and pricing information, including attributes such as make or company name, fuel type, aspiration, body style, drive wheel, engine location, wheelbase, dimensions, curb weight, engine size, horsepower, city and highway fuel consumption, and price. The dataset does not contain full real-world resale variables such as vehicle age, mileage, accident history, service history, condition reports, or location. Therefore, the proposed system must be treated as a specification-based first-level valuation and decision-support tool rather than a complete market valuation system.

Due to increased demand, Mutuka Automotive wants to develop a data-driven vehicle valuation and decision-support system. The system should not only estimate vehicle value, but also help the company categorise vehicles into meaningful valuation groups and identify cases that should be referred for manual review rather than automated pricing.

The proposed system may eventually be integrated into the company's online platform via an API to provide preliminary valuations for prospective sellers. However, before deployment, the company needs evidence that the model is accurate enough for practical use and reliable within clearly defined limits. The company also needs to understand which types of vehicles can be priced with reasonable confidence and which cases require human review.

As a data science consultant, you have been tasked with analysing the supplied dataset, developing and comparing modelling approaches, identifying meaningful vehicle segments or valuation categories, and recommending how Mutuka Automotive should use or limit the model in practice **('car_pricing_datasets.csv')**.

Source: Matanga, N. Y. (2026).

1.1 Clearly define the objective of the study. Your objective must explain the business problem, the analytical task, and what would count as a successful solution. You should identify measurable performance indicators and explain how these indicators relate to Mutuka Automotive's need for reliable vehicle valuation and decision support.

(5 Marks)

1.2 Assess the integrity of the dataset and address any data-quality issues identified. Your analysis should consider missing values, duplicate records, inconsistent categories, incorrect data types, unrealistic or extreme values, and any other issues that may affect modelling. Clearly justify each data-handling decision and explain how poor data quality could affect the final valuation system.

(8 Marks)

1.3 Use appropriate visualisations and summary statistics to explore the relevant attributes. Your EDA should examine distributions, relationships between variables, potential predictors of price or valuation category, and any patterns that may influence model choice. You must interpret the results rather than simply displaying graphs.

(12 Marks)

End of Question 1

# 4. Question 2

## Question 2 25 Marks

From the cleaned dataset, you are required to move beyond basic EDA by identifying meaningful vehicle groups and designing categories that can support business decisions.

2.1 Use analytical reasoning and/or clustering to identify meaningful groups of vehicles in the dataset. The segmentation should be based on variables available in the dataset, such as make or company name derived from CarName, fuel type, aspiration, body style, drive wheel, engine location, curb weight, engine size, horsepower, fuel consumption, dimensions, price, or other relevant specification variables. You must justify the features used for segmentation and explain what the resulting groups mean in the vehicle resale context.

(10 Marks)

2.2 Create meaningful price bands or business decision categories that could be used by Mutuka Automotive. For example, vehicles may be grouped as budget, mid-range, high-value, and premium, or as automatic valuation, manual review, and high-risk review. You must justify how the categories were created and discuss whether the categories are balanced, useful, and realistic.

(9 Marks)

2.3 Explain how the identified vehicle segments and decision categories could help Mutuka Automotive. Your discussion should show how segmentation can support pricing strategy, negotiation, manual inspection, risk identification, or prioritisation of vehicles.

(6 Marks)

End of Question 2

# 5. Question 3

## Question 3 30 Marks

Using the cleaned and prepared dataset, develop and evaluate a modelling solution. The model should support the valuation task and, where relevant, the decision categories designed in Question 2.

3.1 Clearly state the nature of the machine learning problem. Depending on your design, this may involve regression, classification, or a combination of both. Identify the target variable or target categories, select relevant attributes, and perform suitable feature transformations such as extracting the company or make from CarName, encoding categorical variables, scaling numerical variables, handling skewed variables, or creating derived features.

(6 Marks)

3.2 Train and compare at least three modelling approaches:

- one simple baseline model, and
- one more flexible model capable of capturing non-linear relationships.

Report on your findings and recommendations. The final choice should be based on predictive performance, interpretability, generalisation, and suitability for Mutuka Automotive's business context.

(10 Marks)

3.3 Evaluate the models using an appropriate validation strategy and relevant metrics. Regression models may be assessed using metrics such as MAE, RMSE, R-squared, or percentage error, while classification models may be assessed using accuracy, precision, recall, F1-score, confusion matrices, or other justified measures. You must report both training and generalisation performance where appropriate.

(7 Marks)

3.4 Analyse where the final model performs well and where it performs poorly. Consider whether the model tends to overestimate or underestimate certain groups of vehicles, such as low-value vehicles, high-value vehicles, high-horsepower vehicles, large-engine vehicles, fuel-efficient vehicles, rare body styles, premium makes, or vehicles with unusual specification combinations. Explain what these errors imply for business use and identify cases where the model should not be trusted without manual review.

(7 Marks)

End of Question 3

# 6. Question 4

## Question 4 20 Marks

Upon completion of the analysis and model development, you are required to prepare a professional presentation for Mutuka Automotive. The presentation is the required output for this question. It must be suitable for both technical and non-technical stakeholders and must show how the work completed in Questions 1 to 3 supports practical business decisions.

The presentation must not simply report that a model was built. It must synthesise the full project: the business objective, data preparation decisions, EDA insights, vehicle segmentation or valuation grouping, model comparison, evaluation results, error analysis, model trust boundaries, risk-based decision rules, and final deployment recommendation.

4.1. In your presentation, briefly summarise the main findings from Questions 1 to 3 and explain why they matter for Mutuka Automotive. This must include the business objective, important data quality decisions, key EDA insights, the vehicle segmentation or valuation grouping used, and the main findings from the model comparison. Marks should be awarded for interpretation and synthesis, not for merely showing code, tables, or graphs.

(4 Marks)

4.2. Based on the model performance, decide whether the final model is accurate enough to be used as a first-level vehicle valuation tool. Define an acceptable prediction-error range for this dataset and business context, such as whether an error of R1 500, R3 000, or R5 000 would be acceptable, or whether a percentage-based error threshold would be more appropriate. The threshold must be justified in relation to vehicle price, negotiation risk, profit implications, and customer trust, using appropriate metrics, validation results, and visualisations.

(4 Marks)

4.3. Present practical decision rules for how Mutuka Automotive should use the model. For example, vehicles with reliable predictions may receive an automated preliminary estimate; vehicles with unusual specification combinations, rare categories, high predicted error, or values outside the training data range may be referred for manual review; and high-value or high-risk vehicles may require additional inspection before a final offer is made. The decision rules must follow logically from the analysis and model results.

(4 Marks)

4.4. Explain the business consequences of the model's errors. Consider the effect of overpricing, underpricing, misclassification of valuation bands, or incorrect automated approval. You must connect the error analysis from Question 3 to practical risks such as financial loss, unfair offers, customer dissatisfaction, reputational damage, or poor negotiation decisions. Clearly identify the types of vehicles or situations where the model should not be trusted without human review, especially because the dataset does not include mileage, age, condition, service history, or accident history.

(4 Marks)

4.5. End the presentation with a clear recommendation to Mutuka Automotive. State whether the model should be used for full automation, partial automation, decision support only, or further development before deployment. The recommendation must be supported by evidence from the EDA, segmentation or valuation grouping, model comparison, evaluation results, error analysis, and risk-based decision framework.

(4 Marks)

End of Question 4